

UNIVERSITY OF DHAKA

Department of Computer Science and Engineering
Faculty of Engineering and Technology (FoET)

CSE 2206 — Microcontroller and Embedded System

Laboratory MCQ Test | 2nd Year, 2nd Semester 2024

Full Marks: 30 Time: 40 Minutes Each Q: 1 Mark SET A

Student Name: Student ID: Section/Group:

Date: Exam Hall: Seat No.:

Instructions to Students

1. This paper contains **30 MCQ questions**. Each question carries **1 mark**. Total: **30 marks**.
2. **Write all answers on the separate MCQ Answer Sheet only**. Do not write answers on this question paper.
3. **Shade/bubble** the correct option (A/B/C/D) clearly. One bubble per question only.
4. Each question has **exactly one** correct answer. Multiple bubbles score zero.
5. **No negative marking**. Unanswered questions receive zero marks.
6. Mobile phones, programmable calculators, and reference materials are strictly prohibited.
7. This paper **must be submitted** with your Answer Sheet at the end of the examination.

OBE: CO1 (PO1, C3 Apply) CO2 (PO2, C4 Analyse) CO3 (PO6, A4 Organise) | Each question is tagged accordingly.

1. To software-debounce the User Button, a student records the SysTick timestamp at the first falling edge and ignores subsequent edges within 50 ms. The timestamp variable is read by the ISR and by the main loop. Which type qualifier is **essential**? [CO2/PO2]

- (A) `volatile uint32_t`
- (B) `register uint32_t`
- (C) `static uint32_t`
- (D) `const uint32_t`

2. A student observes `RXNE=1` in `USART2->SR`, then immediately enables the `RXNE` interrupt via `CR1 |= USART_CR1_RXNEIE` **without first reading DR**. What happens? [CO2/PO2]

- (A) Nothing unusual; the interrupt arms correctly and fires on the next received byte
- (B) A HardFault occurs because interrupts cannot be enabled while SR flags are pending
- (C) The `USART2` ISR fires **immediately** upon enabling `RXNEIE` because the `RXNE` flag is already pending
- (D) `RXNE` is automatically cleared when `RXNEIE` is set, so no ISR fires

3. Compare two polling transmit strategies:

Strategy P: wait for `TXE` flag, then write to `DR`.

Strategy Q: write to `DR`, then wait for `TC` flag.

What is the key **side effect** of Strategy Q? [CO2/PO2]

-
- (A) The TC flag cannot be polled; it is only available via interrupt
(B) Strategy Q skips the first byte of every new transmission burst
(C) Strategy Q returns before the byte reaches the shift register, causing data loss
(D) Strategy Q blocks for an extra full byte-time *after* the data has left the shift register, unnecessarily reducing maximum throughput
4. A Dhaka-based startup deploys IoT air-quality sensors using STM32 nodes. Each node must transmit 200-byte packets every 500 ms over UART to a LoRa radio while also sampling sensors. Which firmware architecture **best** keeps the CPU free for sensor work? [CO3/PO6]
- (A) DMA TX in normal mode triggered once per 500 ms; CPU enters sleep between transfers
(B) UART TX interrupt per byte with a transmit ring buffer drained by the TXEIE ISR
(C) Bit-bang UART implemented entirely inside a SysTick ISR
(D) Polling TXE in a busy-wait loop for each of the 200 bytes
5. Why must **head** and **tail** be declared **volatile** when shared between the UART ISR and the main loop? [CO2/PO2]
- (A) **volatile** is required for all variables placed in SRAM
(B) **volatile** makes the read-modify-write of the index atomic
(C) Without **volatile**, the compiler may cache the variable in a CPU register and the main loop will *never observe* updates written by the ISR
(D) **volatile** prevents the NVIC from masking the UART interrupt
6. Which path correctly describes the STM32F446 **clock tree** from the HSI oscillator to the USART2 peripheral? [CO1/PO1]
- (A) HSI → RCC → AHB1 → USART2
(B) HSI → SYSCLK → APB2 Prescaler → USART2
(C) HSI → PLL → SYSCLK → AHB Prescaler → APB1 Prescaler → USART2
(D) HSI → PLL → AHB2 → APB1 → USART2
7. The **OSPEEDR** field for a GPIO pin is set to 11 (Very High Speed). What is the **primary physical effect** of this setting? [CO1/PO1]
- (A) The pin switches automatically from push-pull to open-drain mode
(B) The peripheral clock frequency for that GPIO port doubles
(C) The slew rate of the output waveform increases, reducing transition time but potentially increasing EMI and current draw
(D) The LED connected to that pin blinks faster
8. A student reads **USART2->SR** and finds the **ORE** (overrun error) flag set. Which sequence of events **most likely caused** this? [CO2/PO2]
- (A) The transmit shift register ran out of data to send
(B) A new byte arrived in the receive data register before the CPU read the previous byte from DR
(C) The USART peripheral was reset while a byte was being transmitted
(D) The baud rate mismatch was so large that frame synchronisation was permanently lost
9. PA2 is USART2_TX. A student writes **GPIOA->AFR[0] |= (6UL<<8)** instead of **(7UL<<8)**. What is the outcome? [CO2/PO2]
-

- (A) PA2 works for TX but RX on PA3 stops working
- (B) PA2 maps to TIM1_CH1N (AF6) instead of USART2_TX (AF7); UART output is absent or corrupt
- (C) A bus fault occurs because AF6 is reserved on PA2
- (D) AF6 and AF7 both route to USART2 on PA2, so the behaviour is identical

10. The User Button B1 is on PC13. A student configures EXTI line 13 for falling-edge detection but **omits the write to** `SYSCFG->EXTICR[3]`. What is the default EXTI13 source, and what is the practical effect? [CO2/PO2]

- (A) Default source is Port C; the ISR fires correctly even without the SYSCFG write
- (B) EXTI13 has no default and is disabled until SYSCFG is explicitly written
- (C) Default source is Port B; the ISR fires on PB13 events instead
- (D) Default source is Port A (PA13); button presses on PC13 will *not* trigger the ISR

11. For USART2 at 115 200 baud with APB1 = 45 MHz, OVER8 = 0, the correct BRR is mantissa = 390, fraction = 10. A student accidentally sets fraction = 0 (writes `BRR = (390 << 4)`). What is the most likely result? [CO2/PO2]

- (A) Baud rate becomes slightly lower ($\approx 112\,179$ baud); communication may still work at short cable lengths but can produce occasional framing errors
- (B) Baud rate doubles to approximately 230 400 baud
- (C) UART fails to initialise; the TXE flag never sets
- (D) The MCU automatically resets due to a BRR configuration error

12. DMA1 Stream 5 is configured for USART2 RX in **circular mode** with `NDTR=64` and `rxBuf[64]`. After 64 bytes are received, what does the DMA controller do **automatically**? [CO2/PO2]

- (A) It raises a DMA configuration error because circular mode cannot be used with a USART peripheral
- (B) It switches to M1AR (double-buffer mode) automatically
- (C) It resets NDTR to 64, rewinds M0AR to `rxBuf[0]`, raises TC, and continues receiving
- (D) It raises a TC interrupt and halts; the CPU must re-enable the stream manually

13. Bangladesh's SDG Goal 9 targets resilient infrastructure. An engineering team builds a solar-powered river-level monitoring system on STM32 nodes. Which design principle **directly** minimises power consumption for long-term deployment? [CO3/PO6]

- (A) Maximising CPU clock speed to 180 MHz at all times
- (B) Using polling loops for all peripherals to simplify firmware
- (C) Using DMA for background transfers and low-power sleep (`__WFI()`) so the CPU wakes only when real work is needed
- (D) Storing firmware in external SPI Flash to reduce on-chip power dissipation

14. Consider the ISR fragment below: [CO2/PO2]

```
void USART2_IRQHandler(void) {
    char c = (char)USART2->DR;
    ring[head] = c;
    head = (head + 1) % BUF_SIZE;
}
```

A student notices that the ORE flag occasionally freezes the UART receive. What **must** be added to this ISR to fix the problem?

- (A) Check the ORE flag and clear it by reading `USART2->DR` when set

- (B) Call `NVIC_DisableIRQ(USART2_IRQn)` at the top of the ISR
- (C) Replace the modulo operation with a bitmask for execution speed
- (D) Increase `BUF_SIZE` to 256 to prevent overrun

15. Inside `EXTI15_10_IRQHandler()`, a student toggles the LED but **forgets** to clear the EXTI pending register. What is the result? [CO2/PO2]

- (A) The ISR executes once and the pending flag auto-clears after the ISR returns
- (B) The NVIC automatically masks the interrupt after 10 consecutive re-entries
- (C) The processor escalates to a HardFault exception after one re-entry
- (D) The ISR re-enters continuously after each return, creating an infinite interrupt loop

16. A greenhouse controller uses GPIO EXTI for soil-moisture thresholds and DMA for ADC logging. The water valve fails to open during heavy logging. Firmware audit: DMA TC ISR at NVIC priority 1; GPIO EXTI ISR at NVIC priority 3. What is the **root cause**? [CO3/PO6]

- (A) Priority level 3 is numerically higher and therefore *lower* in preemption ability; the DMA ISR prevents the moisture EXTI ISR from executing in time
- (B) The ADC DMA fills the ring buffer and overwrites the threshold comparison variable
- (C) GPIO EXTI interrupts are hardware-disabled while DMA is running
- (D) DMA transfers corrupt the GPIO output data register

17. Before accessing any register of GPIOA, a student writes `RCC->AHB1ENR |= RCC_AHB1ENR_GPIOAEN`. If this line is **omitted**, what is the most likely hardware behaviour? [CO1/PO1]

- (A) GPIOA defaults to input mode and still works correctly for reading
- (B) The processor hangs at reset because GPIO is required for the boot sequence
- (C) GPIOA registers read as 0x00000000 and writes are silently discarded
- (D) A HardFault exception is raised immediately on the first GPIOA access

18. Which statement about RS-485 versus UART (RS-232) is **correct** in the context of STM32 communications? [CO2/PO2]

- (A) RS-232 operates at up to 10 Mbps; RS-485 is limited to 115 200 baud
- (B) RS-485 is single-ended; RS-232 uses differential signalling
- (C) RS-485 supports multi-drop bus networks (many nodes on one twisted pair); RS-232 is point-to-point only
- (D) STM32 USART peripherals natively generate RS-485 voltage levels without any external transceiver

19. Which single C statement **correctly** clears the EXTI pending flag for line 13 inside an ISR? (The EXTI PR register is write-1-to-clear.) [CO2/PO2]

- (A) `EXTI->PR = (1UL<<13);`
- (B) `NVIC_ClearPendingIRQ(EXTI15_10_IRQn);`
- (C) `EXTI->PR &= ~(1UL<<13);`
- (D) `EXTI->IMR &= ~(1UL<<13);`

20. A student must configure PA5 as a general-purpose push-pull output. Which statement about the required register writes is **correct**? [CO1/PO1]

- (A) Only `ODR[5]=1` needs to be set to activate the output

- (B) `MODER[11:10]=01` must be written; the reset defaults of `OTYPER` (push-pull) and `PUPDR` (no pull) are already correct
- (C) Both `MODER[11:10]=10` and an `AFR` entry must be written
- (D) Both `MODER[11:10]=01` and `OTYPER[5]=1` must be written

21. USART2 ISR has NVIC preemption priority 0; EXTI15_10 ISR has preemption priority 2. The User Button is pressed while the USART2 ISR is executing. What happens? [CO2/PO2]

- (A) The EXTI15_10 ISR is pended and waits until the USART2 ISR completes, because a higher priority number means lower preemption ability on Cortex-M4
- (B) A fault occurs because two ISRs cannot be pending simultaneously in the NVIC
- (C) The EXTI15_10 ISR immediately preempts USART2 ISR because button events are hardware-urgent
- (D) Both ISRs execute simultaneously on the dual-issue Cortex-M4 pipeline

22. A team designs a fault-tolerant industrial control network spanning 100 m with 8 nodes on the same wired bus in an electrically noisy factory. Which STM32 communication protocol is **most appropriate**? [CO3/PO6]

- (A) UART RS-232 with star topology
- (B) SPI with daisy-chain topology
- (C) I²C at 400 kHz Fast Mode
- (D) CAN bus (bxCAN/fdCAN) with differential signalling

23. `GPIOA->MODER` is a 32-bit register with 2 bits per pin. What is the correct C expression to set `PA8` as Alternate Function mode without changing any other bits? [CO1/PO1]

- (A) `GPIOA->MODER = (2UL << 16);`
- (B) `GPIOA->MODER &= ~(3UL<<16); GPIOA->MODER |= (2UL<<16);`
- (C) `GPIOA->MODER |= (2UL<<8);`
- (D) `GPIOA->MODER &= ~(3UL<<8); GPIOA->MODER |= (2UL<<8);`

24. A student enables DMA1 Stream6 for USART2 TX but selects **Channel 2** instead of the correct **Channel 4**. DMA is enabled and `USART2->CR3 |= USART_CR3_DMAT`. What is the most likely symptom? [CO2/PO2]

- (A) The stream is enabled but no peripheral request arrives on Channel 2 from USART2; the transfer never starts and TC is never asserted
- (B) A DMA error interrupt fires immediately upon enabling the stream
- (C) USART2 transmits random data from an unrelated memory region
- (D) The DMA transfer completes instantly with zero bytes sent

25. After a system reset on STM32F446, which clock source is active **before** any firmware initialises the RCC? [CO1/PO1]

- (A) HSE (High-Speed External) at 8 MHz
- (B) LSI (Low-Speed Internal RC) at 32 kHz
- (C) HSI (High-Speed Internal RC oscillator) at 16 MHz
- (D) PLL locked to 180 MHz

26. A student configures USART2 with APB1 clock = 45 MHz, `OVER8 = 0`. They write `USART2->BRR = 0x1A1` (decimal 417). What is the **actual baud rate** produced? (*Hint: actual baud = f_{CK}/BRR*)

[CO2/PO2]

- (A) 115 200 baud
- (B) Approximately 107 914 baud
- (C) Approximately 9 600 baud
- (D) Approximately 230 400 baud

27. A student uses `HAL_UART_Receive_IT(&huart2, &rxByte, 1)` and re-arms it in the callback. At 115 200 baud one byte arrives every $\approx 87 \mu\text{s}$. The callback takes $200 \mu\text{s}$. What is the **most likely** consequence? [CO2/PO2]

- (A) One or more incoming bytes are lost while the callback executes, because RXNE fires again before `HAL_UART_Receive_IT` is re-armed
- (B) HAL automatically queues up to 16 bytes in the USART FIFO
- (C) The next byte is safely stored in a hardware double-buffer inside the USART peripheral
- (D) The MCU resets due to a watchdog timeout triggered by the long callback

28. The STM32F446RE Nucleo-64 contains an ARM Cortex-M4 core. GPIO peripherals are connected to the processor via which bus? [CO1/PO1]

- (A) APB2 (Advanced Peripheral Bus 2)
- (B) AHB1 (Advanced High-performance Bus 1)
- (C) APB1 (Advanced Peripheral Bus 1)
- (D) AHB2 (Advanced High-performance Bus 2)

29. Two methods set PA5 HIGH then LOW:

Method X: `GPIOA->ODR |= (1<<5) ... GPIOA->ODR &= ~(1<<5)`

Method Y: `GPIOA->BSRR = (1<<5) ... GPIOA->BSRR = (1<<21)`

Why is Method Y preferred inside an ISR? [CO1/PO1]

- (A) Method X requires a non-atomic read-modify-write; an interrupt between the read and write can corrupt ODR bits belonging to other pins
- (B) Method Y enables the hardware to automatically toggle PA5 at a fixed frequency
- (C) Method X uses more Flash because the compiler cannot optimise bitwise OR/AND on ODR
- (D) Method Y executes in a single machine cycle; Method X takes two

30. A ring buffer uses `BUF_SIZE=64` and indices `volatile uint8_t head=0, tail=0;`. The ISR increments `head`; main increments `tail`. How many bytes can the buffer hold **before it is full**? [CO2/PO2]

- (A) 63 bytes — one slot is reserved so the full condition `(head+1)%SIZE == tail` can be distinguished from empty
- (B) 64 bytes
- (C) 62 bytes
- (D) 32 bytes (half capacity is the recommended safe limit)

— End of Question Paper —

Total Questions: 30 | Total Marks: 30 | Submit this paper with your Answer Sheet

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CSE 2206 — Microcontroller and Embedded System

MCQ ANSWER SHEET SET A

2nd Year, 2nd Semester 2024 | Full Marks: 30 | Time: 40 Minutes

Student Name: Student ID: Section/Group:

Date: Exam Hall: Seat No.:

Instructions: Shade **one** bubble per question using a dark pen or pencil. Do not use ticks, crosses, or circles. Write your answers *here only* — not on the question paper. Erased or ambiguous answers score zero.

Answer Grid — Shade the correct bubble

Q#	A	B	C	D	Q#	A	B	C	D
1	☐	☐	☐	☐	16	☐	☐	☐	☐
2	☐	☐	☐	☐	17	☐	☐	☐	☐
3	☐	☐	☐	☐	18	☐	☐	☐	☐
4	☐	☐	☐	☐	19	☐	☐	☐	☐
5	☐	☐	☐	☐	20	☐	☐	☐	☐
6	☐	☐	☐	☐	21	☐	☐	☐	☐
7	☐	☐	☐	☐	22	☐	☐	☐	☐
8	☐	☐	☐	☐	23	☐	☐	☐	☐
9	☐	☐	☐	☐	24	☐	☐	☐	☐
10	☐	☐	☐	☐	25	☐	☐	☐	☐
11	☐	☐	☐	☐	26	☐	☐	☐	☐
12	☐	☐	☐	☐	27	☐	☐	☐	☐
13	☐	☐	☐	☐	28	☐	☐	☐	☐
14	☐	☐	☐	☐	29	☐	☐	☐	☐
15	☐	☐	☐	☐	30	☐	☐	☐	☐

For Examiner Use Only

CO1 (PO1)	CO2 (PO2)	CO3 (PO6)	Total /30	Percentage
/8	/18	/4	/30	%

Examiner Name: **Signature:**

ANSWER KEY — Examiner Copy Only. Remove before printing student sheets.

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans		
1	A	2	C	3	D	4	A	5	C	6	C
7	C	8	B	9	B	10	D	11	A	12	C
13	C	14	A	15	D	16	A	17	C	18	C
19	A	20	B	21	A	22	D	23	B	24	A
25	C	26	B	27	A	28	B	29	A	30	A

— End of Answer Sheet —

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CSE 2206 — Microcontroller and Embedded System

Laboratory MCQ Test | 2nd Year, 2nd Semester 2024

Full Marks: 30 Time: 40 Minutes Each Q: 1 Mark **SET B**

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1. A greenhouse controller uses GPIO EXTI for soil-moisture thresholds and DMA for ADC logging. The water valve fails to open during heavy logging. Firmware audit: DMA TC ISR at NVIC priority 1; GPIO EXTI ISR at NVIC priority 3. What is the **root cause**? [CO3/PO6]

- (A) DMA transfers corrupt the GPIO output data register
- (B) GPIO EXTI interrupts are hardware-disabled while DMA is running
- (C) Priority level 3 is numerically higher and therefore *lower* in preemption ability; the DMA ISR prevents the moisture EXTI ISR from executing in time
- (D) The ADC DMA fills the ring buffer and overwrites the threshold comparison variable

2. `GPIOA->MODER` is a 32-bit register with 2 bits per pin. What is the correct C expression to set `PA8` as Alternate Function mode without changing any other bits? [CO1/PO1]

- (A) `GPIOA->MODER &= ~(3UL<<8); GPIOA->MODER |= (2UL<<8);`
- (B) `GPIOA->MODER &= ~(3UL<<16); GPIOA->MODER |= (2UL<<16);`
- (C) `GPIOA->MODER |= (2UL<<8);`
- (D) `GPIOA->MODER = (2UL << 16);`

3. Why must **head** and **tail** be declared **volatile** when shared between the UART ISR and the main loop? [CO2/PO2]

- (A) **volatile** is required for all variables placed in SRAM
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- (D) **volatile** makes the read-modify-write of the index atomic

4. Which single C statement **correctly** clears the EXTI pending flag for line 13 inside an ISR? (The EXTI PR register is write-1-to-clear.) [CO2/PO2]

- (A) `EXTI->PR = (1UL<<13);`
- (B) `EXTI->IMR &= ~(1UL<<13);`
- (C) `NVIC_ClearPendingIRQ(EXTI15_10_IRQn);`
- (D) `EXTI->PR &= ~(1UL<<13);`

5. Before accessing any register of GPIOA, a student writes `RCC->AHB1ENR |= RCC_AHB1ENR_GPIOAEN`. If this line is **omitted**, what is the most likely hardware behaviour? [CO1/PO1]

- (A) GPIOA registers read as 0x00000000 and writes are silently discarded
- (B) A HardFault exception is raised immediately on the first GPIOA access
- (C) The processor hangs at reset because GPIO is required for the boot sequence
- (D) GPIOA defaults to input mode and still works correctly for reading

6. A student enables DMA1 Stream6 for USART2 TX but selects **Channel 2** instead of the correct **Channel 4**. DMA is enabled and `USART2->CR3 |= USART_CR3_DMAT`. What is the most likely symptom? [CO2/PO2]

- (A) The DMA transfer completes instantly with zero bytes sent
- (B) The stream is enabled but no peripheral request arrives on Channel 2 from USART2; the transfer never starts and TC is never asserted
- (C) USART2 transmits random data from an unrelated memory region
- (D) A DMA error interrupt fires immediately upon enabling the stream

7. Two methods set PA5 HIGH then LOW:

Method X: `GPIOA->ODR |= (1<<5) ... GPIOA->ODR &= ~(1<<5)`

Method Y: `GPIOA->BSRR = (1<<5) ... GPIOA->BSRR = (1<<21)`

Why is Method Y preferred inside an ISR? [CO1/PO1]

- (A) Method X requires a non-atomic read-modify-write; an interrupt between the read and write can corrupt ODR bits belonging to other pins
- (B) Method Y executes in a single machine cycle; Method X takes two
- (C) Method Y enables the hardware to automatically toggle PA5 at a fixed frequency
- (D) Method X uses more Flash because the compiler cannot optimise bitwise OR/AND on ODR

8. A student observes `RXNE=1` in `USART2->SR`, then immediately enables the RXNE interrupt via `CR1 |= USART_CR1_RXNEIE` **without first reading DR**. What happens? [CO2/PO2]

- (A) A HardFault occurs because interrupts cannot be enabled while SR flags are pending
- (B) Nothing unusual; the interrupt arms correctly and fires on the next received byte
- (C) RXNE is automatically cleared when RXNEIE is set, so no ISR fires
- (D) The USART2 ISR fires **immediately** upon enabling RXNEIE because the RXNE flag is already pending

9. For USART2 at 115 200 baud with $APB1 = 45\text{ MHz}$, $OVER8 = 0$, the correct BRR is mantissa = 390, fraction = 10. A student accidentally sets fraction = 0 (writes $BRR = (390 \ll 4)$). What is the most likely result? [CO2/PO2]

- (A) UART fails to initialise; the TXE flag never sets
- (B) Baud rate doubles to approximately 230 400 baud
- (C) The MCU automatically resets due to a BRR configuration error
- (D) Baud rate becomes slightly lower ($\approx 112\,179$ baud); communication may still work at short cable lengths but can produce occasional framing errors

10. Consider the ISR fragment below: [CO2/PO2]

```
void USART2_IRQHandler(void) {
    char c = (char)USART2->DR;
    ring[head] = c;
    head = (head + 1) % BUF_SIZE;
}
```

A student notices that the ORE flag occasionally freezes the UART receive. What **must** be added to this ISR to fix the problem?

- (A) Check the ORE flag and clear it by reading `USART2->DR` when set
- (B) Replace the modulo operation with a bitmask for execution speed
- (C) Call `NVIC_DisableIRQ(USART2_IRQn)` at the top of the ISR
- (D) Increase `BUF_SIZE` to 256 to prevent overrun

11. A team designs a fault-tolerant industrial control network spanning 100 m with 8 nodes on the same wired bus in an electrically noisy factory. Which STM32 communication protocol is **most appropriate**? [CO3/PO6]

- (A) I²C at 400 kHz Fast Mode
- (B) CAN bus (bxCAN/fdCAN) with differential signalling
- (C) UART RS-232 with star topology
- (D) SPI with daisy-chain topology

12. USART2 ISR has NVIC preemption priority 0; EXTI15_10 ISR has preemption priority 2. The User Button is pressed while the USART2 ISR is executing. What happens? [CO2/PO2]

- (A) The EXTI15_10 ISR immediately preempts USART2 ISR because button events are hardware-urgent
- (B) The EXTI15_10 ISR is pended and waits until the USART2 ISR completes, because a higher priority number means lower preemption ability on Cortex-M4
- (C) A fault occurs because two ISRs cannot be pending simultaneously in the NVIC
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13. A student configures USART2 with $APB1\text{ clock} = 45\text{ MHz}$, $OVER8 = 0$. They write `USART2->BRR = 0x1A1` (decimal 417). What is the **actual baud rate** produced? (*Hint: $\text{actual baud} = f_{CK}/BRR$*) [CO2/PO2]

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- (B) Approximately 230 400 baud
- (C) Approximately 9 600 baud
- (D) Approximately 107 914 baud

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- (A) It resets NDTR to 64, rewinds M0AR to `rxBuf[0]`, raises TC, and continues receiving
- (B) It raises a TC interrupt and halts; the CPU must re-enable the stream manually
- (C) It switches to M1AR (double-buffer mode) automatically
- (D) It raises a DMA configuration error because circular mode cannot be used with a USART peripheral

15. Bangladesh's SDG Goal 9 targets resilient infrastructure. An engineering team builds a solar-powered river-level monitoring system on STM32 nodes. Which design principle **directly** minimises power consumption for long-term deployment? [CO3/PO6]

- (A) Using DMA for background transfers and low-power sleep (`__WFI()`) so the CPU wakes only when real work is needed
- (B) Maximising CPU clock speed to 180 MHz at all times
- (C) Storing firmware in external SPI Flash to reduce on-chip power dissipation
- (D) Using polling loops for all peripherals to simplify firmware

16. A student uses `HAL_UART_Receive_IT(&huart2, &rxByte, 1)` and re-arms it in the callback. At 115 200 baud one byte arrives every $\approx 87 \mu\text{s}$. The callback takes $200 \mu\text{s}$. What is the **most likely** consequence? [CO2/PO2]

- (A) One or more incoming bytes are lost while the callback executes, because RXNE fires again before `HAL_UART_Receive_IT` is re-armed
- (B) The MCU resets due to a watchdog timeout triggered by the long callback
- (C) HAL automatically queues up to 16 bytes in the USART FIFO
- (D) The next byte is safely stored in a hardware double-buffer inside the USART peripheral

17. To software-debounce the User Button, a student records the SysTick timestamp at the first falling edge and ignores subsequent edges within 50 ms. The timestamp variable is read by the ISR and by the main loop. Which type qualifier is **essential**? [CO2/PO2]

- (A) `volatile uint32_t`
- (B) `static uint32_t`
- (C) `register uint32_t`
- (D) `const uint32_t`

18. A ring buffer uses `BUF_SIZE=64` and indices `volatile uint8_t head=0, tail=0;`. The ISR increments `head`; main increments `tail`. How many bytes can the buffer hold **before it is full**? [CO2/PO2]

- (A) 32 bytes (half capacity is the recommended safe limit)
- (B) 63 bytes — one slot is reserved so the full condition `(head+1)%SIZE == tail` can be distinguished from empty
- (C) 62 bytes
- (D) 64 bytes

19. A Dhaka-based startup deploys IoT air-quality sensors using STM32 nodes. Each node must transmit 200-byte packets every 500 ms over UART to a LoRa radio while also sampling sensors. Which firmware architecture **best** keeps the CPU free for sensor work? [CO3/PO6]

- (A) DMA TX in normal mode triggered once per 500 ms; CPU enters sleep between transfers
- (B) Polling TXE in a busy-wait loop for each of the 200 bytes
- (C) UART TX interrupt per byte with a transmit ring buffer drained by the TXEIE ISR

(D) Bit-bang UART implemented entirely inside a SysTick ISR

20. Inside `EXTI15_10_IRQHandler()`, a student toggles the LED but **forgets** to clear the EXTI pending register. What is the result? [CO2/PO2]

- (A) The NVIC automatically masks the interrupt after 10 consecutive re-entries
- (B) The ISR executes once and the pending flag auto-clears after the ISR returns
- (C) The processor escalates to a HardFault exception after one re-entry
- (D) The ISR re-enters continuously after each return, creating an infinite interrupt loop

21. A student reads `USART2->SR` and finds the **ORE** (overrun error) flag set. Which sequence of events **most likely caused** this? [CO2/PO2]

- (A) The transmit shift register ran out of data to send
- (B) The USART peripheral was reset while a byte was being transmitted
- (C) The baud rate mismatch was so large that frame synchronisation was permanently lost
- (D) A new byte arrived in the receive data register before the CPU read the previous byte from DR

22. The STM32F446RE Nucleo-64 contains an ARM Cortex-M4 core. GPIO peripherals are connected to the processor via which bus? [CO1/PO1]

- (A) AHB1 (Advanced High-performance Bus 1)
- (B) APB1 (Advanced Peripheral Bus 1)
- (C) AHB2 (Advanced High-performance Bus 2)
- (D) APB2 (Advanced Peripheral Bus 2)

23. PA2 is `USART2_TX`. A student writes `GPIOA->AFR[0] |= (6UL<<8)` instead of `(7UL<<8)`. What is the outcome? [CO2/PO2]

- (A) PA2 works for TX but RX on PA3 stops working
- (B) A bus fault occurs because AF6 is reserved on PA2
- (C) AF6 and AF7 both route to USART2 on PA2, so the behaviour is identical
- (D) PA2 maps to `TIM1_CH1N` (AF6) instead of `USART2_TX` (AF7); UART output is absent or corrupt

24. The User Button B1 is on PC13. A student configures EXTI line 13 for falling-edge detection but **omits the write to** `SYSCFG->EXTICR[3]`. What is the default EXTI13 source, and what is the practical effect? [CO2/PO2]

- (A) Default source is Port A (PA13); button presses on PC13 will *not* trigger the ISR
- (B) Default source is Port C; the ISR fires correctly even without the SYSCFG write
- (C) Default source is Port B; the ISR fires on PB13 events instead
- (D) EXTI13 has no default and is disabled until SYSCFG is explicitly written

25. After a system reset on STM32F446, which clock source is active **before** any firmware initialises the RCC? [CO1/PO1]

- (A) HSI (High-Speed Internal RC oscillator) at 16 MHz
- (B) PLL locked to 180 MHz
- (C) LSI (Low-Speed Internal RC) at 32 kHz
- (D) HSE (High-Speed External) at 8 MHz

26. Which path correctly describes the STM32F446 **clock tree** from the HSI oscillator to the USART2 peripheral? [CO1/PO1]

- (A) HSI → RCC → AHB1 → USART2
- (B) HSI → PLL → AHB2 → APB1 → USART2
- (C) HSI → SYSCLK → APB2 Prescaler → USART2
- (D) HSI → PLL → SYSCLK → AHB Prescaler → APB1 Prescaler → USART2

27. Compare two polling transmit strategies:

Strategy P: wait for **TXE** flag, then write to **DR**.

Strategy Q: write to **DR**, then wait for **TC** flag.

What is the key **side effect** of Strategy Q? [CO2/PO2]

- (A) Strategy Q returns before the byte reaches the shift register, causing data loss
- (B) Strategy Q skips the first byte of every new transmission burst
- (C) Strategy Q blocks for an extra full byte-time *after* the data has left the shift register, unnecessarily reducing maximum throughput
- (D) The TC flag cannot be polled; it is only available via interrupt

28. Which statement about RS-485 versus UART (RS-232) is **correct** in the context of STM32 communications? [CO2/PO2]

- (A) STM32 USART peripherals natively generate RS-485 voltage levels without any external transceiver
- (B) RS-232 operates at up to 10 Mbps; RS-485 is limited to 115 200 baud
- (C) RS-485 supports multi-drop bus networks (many nodes on one twisted pair); RS-232 is point-to-point only
- (D) RS-485 is single-ended; RS-232 uses differential signalling

29. The **OSPEEDR** field for a GPIO pin is set to 11 (Very High Speed). What is the **primary physical effect** of this setting? [CO1/PO1]

- (A) The pin switches automatically from push-pull to open-drain mode
- (B) The peripheral clock frequency for that GPIO port doubles
- (C) The slew rate of the output waveform increases, reducing transition time but potentially increasing EMI and current draw
- (D) The LED connected to that pin blinks faster

30. A student must configure PA5 as a general-purpose push-pull output. Which statement about the required register writes is **correct**? [CO1/PO1]

- (A) Both **MODER[11:10]=01** and **OTYPER[5]=1** must be written
- (B) **MODER[11:10]=01** must be written; the reset defaults of OTYPER (push-pull) and PUPDR (no pull) are already correct
- (C) Only **ODR[5]=1** needs to be set to activate the output
- (D) Both **MODER[11:10]=10** and an AFR entry must be written

— End of Question Paper —

Total Questions: 30 | Total Marks: 30 | Submit this paper with your Answer Sheet

UNIVERSITY OF DHAKA

Department of Computer Science and Engineering

CSE 2206 — Microcontroller and Embedded System

MCQ ANSWER SHEET SET B

2nd Year, 2nd Semester 2024 | Full Marks: 30 | Time: 40 Minutes

Student Name: Student ID: Section/Group:

Date: Exam Hall: Seat No.:

Instructions: Shade **one** bubble per question using a dark pen or pencil. Do not use ticks, crosses, or circles. Write your answers *here only* — not on the question paper. Erased or ambiguous answers score zero.

Answer Grid — Shade the correct bubble

Q#	A	B	C	D	Q#	A	B	C	D
1	☐	☐	☐	☐	16	☐	☐	☐	☐
2	☐	☐	☐	☐	17	☐	☐	☐	☐
3	☐	☐	☐	☐	18	☐	☐	☐	☐
4	☐	☐	☐	☐	19	☐	☐	☐	☐
5	☐	☐	☐	☐	20	☐	☐	☐	☐
6	☐	☐	☐	☐	21	☐	☐	☐	☐
7	☐	☐	☐	☐	22	☐	☐	☐	☐
8	☐	☐	☐	☐	23	☐	☐	☐	☐
9	☐	☐	☐	☐	24	☐	☐	☐	☐
10	☐	☐	☐	☐	25	☐	☐	☐	☐
11	☐	☐	☐	☐	26	☐	☐	☐	☐
12	☐	☐	☐	☐	27	☐	☐	☐	☐
13	☐	☐	☐	☐	28	☐	☐	☐	☐
14	☐	☐	☐	☐	29	☐	☐	☐	☐
15	☐	☐	☐	☐	30	☐	☐	☐	☐

For Examiner Use Only

CO1 (PO1)	CO2 (PO2)	CO3 (PO6)	Total /30	Percentage
/8	/18	/4	/30	%

Examiner Name: **Signature:**

ANSWER KEY — Examiner Copy Only. Remove before printing student sheets.

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans		
1	C	2	B	3	B	4	A	5	A	6	B
7	A	8	D	9	D	10	A	11	B	12	B
13	D	14	A	15	A	16	A	17	A	18	B
19	A	20	D	21	D	22	A	23	D	24	A
25	A	26	D	27	C	28	C	29	C	30	B

— End of Answer Sheet —

UNIVERSITY OF DHAKA

Department of Computer Science and Engineering

Faculty of Engineering and Technology (FoET)

CSE 2206 — Microcontroller and Embedded System

Laboratory MCQ Test | 2nd Year, 2nd Semester 2024

Full Marks: 30 Time: 40 Minutes Each Q: 1 Mark **SET C**

Student Name: Student ID: Section/Group:

Date: Exam Hall: Seat No.:

Instructions to Students

1. This paper contains **30 MCQ questions**. Each question carries **1 mark**. Total: **30 marks**.
2. **Write all answers on the separate MCQ Answer Sheet only**. Do not write answers on this question paper.
3. **Shade/bubble** the correct option (A/B/C/D) clearly. One bubble per question only.
4. Each question has **exactly one** correct answer. Multiple bubbles score zero.
5. **No negative marking**. Unanswered questions receive zero marks.
6. Mobile phones, programmable calculators, and reference materials are strictly prohibited.
7. This paper **must be submitted** with your Answer Sheet at the end of the examination.

OBE: CO1 (PO1, C3 Apply) CO2 (PO2, C4 Analyse) CO3 (PO6, A4 Organise) | Each question is tagged accordingly.

1. Why must **head** and **tail** be declared **volatile** when shared between the UART ISR and the main loop? [CO2/PO2]

- (A) **volatile** makes the read-modify-write of the index atomic
- (B) **volatile** is required for all variables placed in SRAM
- (C) **volatile** prevents the NVIC from masking the UART interrupt
- (D) Without **volatile**, the compiler may cache the variable in a CPU register and the main loop will *never observe* updates written by the ISR

2. A student observes **RXNE=1** in USART2->SR, then immediately enables the RXNE interrupt via **CR1 |= USART_CR1_RXNEIE** without first reading **DR**. What happens? [CO2/PO2]

- (A) RXNE is automatically cleared when RXNEIE is set, so no ISR fires
- (B) The USART2 ISR fires **immediately** upon enabling RXNEIE because the RXNE flag is already pending
- (C) Nothing unusual; the interrupt arms correctly and fires on the next received byte
- (D) A HardFault occurs because interrupts cannot be enabled while SR flags are pending

3. The **OSPEEDR** field for a GPIO pin is set to 11 (Very High Speed). What is the **primary physical effect** of this setting? [CO1/PO1]

-
- (A) The slew rate of the output waveform increases, reducing transition time but potentially increasing EMI and current draw
- (B) The pin switches automatically from push-pull to open-drain mode
- (C) The LED connected to that pin blinks faster
- (D) The peripheral clock frequency for that GPIO port doubles
4. A student reads `USART2->SR` and finds the `ORE` (overrun error) flag set. Which sequence of events **most likely** caused this? [CO2/PO2]
- (A) The USART peripheral was reset while a byte was being transmitted
- (B) The baud rate mismatch was so large that frame synchronisation was permanently lost
- (C) The transmit shift register ran out of data to send
- (D) A new byte arrived in the receive data register before the CPU read the previous byte from DR
5. After a system reset on STM32F446, which clock source is active **before** any firmware initialises the RCC? [CO1/PO1]
- (A) HSE (High-Speed External) at 8 MHz
- (B) HSI (High-Speed Internal RC oscillator) at 16 MHz
- (C) PLL locked to 180 MHz
- (D) LSI (Low-Speed Internal RC) at 32 kHz
6. Which statement about RS-485 versus UART (RS-232) is **correct** in the context of STM32 communications? [CO2/PO2]
- (A) RS-232 operates at up to 10 Mbps; RS-485 is limited to 115 200 baud
- (B) RS-485 supports multi-drop bus networks (many nodes on one twisted pair); RS-232 is point-to-point only
- (C) STM32 USART peripherals natively generate RS-485 voltage levels without any external transceiver
- (D) RS-485 is single-ended; RS-232 uses differential signalling
7. A greenhouse controller uses GPIO EXTI for soil-moisture thresholds and DMA for ADC logging. The water valve fails to open during heavy logging. Firmware audit: DMA TC ISR at NVIC priority 1; GPIO EXTI ISR at NVIC priority 3. What is the **root cause**? [CO3/PO6]
- (A) GPIO EXTI interrupts are hardware-disabled while DMA is running
- (B) The ADC DMA fills the ring buffer and overwrites the threshold comparison variable
- (C) Priority level 3 is numerically higher and therefore *lower* in preemption ability; the DMA ISR prevents the moisture EXTI ISR from executing in time
- (D) DMA transfers corrupt the GPIO output data register
8. Before accessing any register of GPIOA, a student writes `RCC->AHB1ENR |= RCC_AHB1ENR_GPIOAEN`. If this line is **omitted**, what is the most likely hardware behaviour? [CO1/PO1]
- (A) The processor hangs at reset because GPIO is required for the boot sequence
- (B) GPIOA defaults to input mode and still works correctly for reading
- (C) GPIOA registers read as 0x00000000 and writes are silently discarded
- (D) A HardFault exception is raised immediately on the first GPIOA access
9. To software-debounce the User Button, a student records the SysTick timestamp at the first falling edge and ignores subsequent edges within 50 ms. The timestamp variable is read by the ISR and by the main loop. Which type qualifier is **essential**? [CO2/PO2]
-

- (A) `volatile uint32_t`
- (B) `const uint32_t`
- (C) `register uint32_t`
- (D) `static uint32_t`

10. Bangladesh's SDG Goal 9 targets resilient infrastructure. An engineering team builds a solar-powered river-level monitoring system on STM32 nodes. Which design principle **directly** minimises power consumption for long-term deployment? [CO3/PO6]

- (A) Using polling loops for all peripherals to simplify firmware
- (B) Storing firmware in external SPI Flash to reduce on-chip power dissipation
- (C) Maximising CPU clock speed to 180 MHz at all times
- (D) Using DMA for background transfers and low-power sleep (`__WFI()`) so the CPU wakes only when real work is needed

11. A student enables DMA1 Stream6 for USART2 TX but selects **Channel 2** instead of the correct **Channel 4**. DMA is enabled and `USART2->CR3 |= USART_CR3_DMAT`. What is the most likely symptom? [CO2/PO2]

- (A) A DMA error interrupt fires immediately upon enabling the stream
- (B) The DMA transfer completes instantly with zero bytes sent
- (C) The stream is enabled but no peripheral request arrives on Channel 2 from USART2; the transfer never starts and TC is never asserted
- (D) USART2 transmits random data from an unrelated memory region

12. Inside `EXTI15_10_IRQHandler()`, a student toggles the LED but **forgets** to clear the EXTI pending register. What is the result? [CO2/PO2]

- (A) The ISR re-enters continuously after each return, creating an infinite interrupt loop
- (B) The processor escalates to a HardFault exception after one re-entry
- (C) The ISR executes once and the pending flag auto-clears after the ISR returns
- (D) The NVIC automatically masks the interrupt after 10 consecutive re-entries

13. PA2 is USART2_TX. A student writes `GPIOA->AFR[0] |= (6UL<<8)` instead of `(7UL<<8)`. What is the outcome? [CO2/PO2]

- (A) AF6 and AF7 both route to USART2 on PA2, so the behaviour is identical
- (B) PA2 maps to TIM1_CH1N (AF6) instead of USART2_TX (AF7); UART output is absent or corrupt
- (C) A bus fault occurs because AF6 is reserved on PA2
- (D) PA2 works for TX but RX on PA3 stops working

14. The STM32F446RE Nucleo-64 contains an ARM Cortex-M4 core. GPIO peripherals are connected to the processor via which bus? [CO1/PO1]

- (A) APB2 (Advanced Peripheral Bus 2)
- (B) APB1 (Advanced Peripheral Bus 1)
- (C) AHB1 (Advanced High-performance Bus 1)
- (D) AHB2 (Advanced High-performance Bus 2)

15. For USART2 at 115 200 baud with $APB1 = 45\text{ MHz}$, $OVER8 = 0$, the correct BRR is mantissa = 390, fraction = 10. A student accidentally sets fraction = 0 (writes `BRR = (390 << 4)`). What is

the most likely result? [CO2/PO2]

- (A) Baud rate doubles to approximately 230 400 baud
- (B) The MCU automatically resets due to a BRR configuration error
- (C) Baud rate becomes slightly lower ($\approx 112\,179$ baud); communication may still work at short cable lengths but can produce occasional framing errors
- (D) UART fails to initialise; the TXE flag never sets

16. A student configures USART2 with APB1 clock = 45 MHz, OVER8 = 0. They write `USART2->BRR = 0x1A1` (decimal 417). What is the **actual baud rate** produced? (*Hint: $\text{actual baud} = f_{CK}/BRR$*) [CO2/PO2]

- (A) Approximately 230 400 baud
- (B) 115 200 baud
- (C) Approximately 107 914 baud
- (D) Approximately 9 600 baud

17. A team designs a fault-tolerant industrial control network spanning 100 m with 8 nodes on the same wired bus in an electrically noisy factory. Which STM32 communication protocol is **most appropriate**? [CO3/PO6]

- (A) SPI with daisy-chain topology
- (B) CAN bus (bxCAN/fdCAN) with differential signalling
- (C) I²C at 400 kHz Fast Mode
- (D) UART RS-232 with star topology

18. Which path correctly describes the STM32F446 **clock tree** from the HSI oscillator to the USART2 peripheral? [CO1/PO1]

- (A) HSI $\rightarrow$ SYSCLK $\rightarrow$ APB2 Prescaler $\rightarrow$ USART2
- (B) HSI $\rightarrow$ PLL $\rightarrow$ AHB2 $\rightarrow$ APB1 $\rightarrow$ USART2
- (C) HSI $\rightarrow$ PLL $\rightarrow$ SYSCLK $\rightarrow$ AHB Prescaler $\rightarrow$ APB1 Prescaler $\rightarrow$ USART2
- (D) HSI $\rightarrow$ RCC $\rightarrow$ AHB1 $\rightarrow$ USART2

19. Two methods set PA5 HIGH then LOW:

Method X: `GPIOA->ODR |= (1<5) ... GPIOA->ODR &= ~(1<5)`

Method Y: `GPIOA->BSRR = (1<5) ... GPIOA->BSRR = (1<21)`

Why is Method Y preferred inside an ISR? [CO1/PO1]

- (A) Method X requires a non-atomic read-modify-write; an interrupt between the read and write can corrupt ODR bits belonging to other pins
- (B) Method X uses more Flash because the compiler cannot optimise bitwise OR/AND on ODR
- (C) Method Y executes in a single machine cycle; Method X takes two
- (D) Method Y enables the hardware to automatically toggle PA5 at a fixed frequency

20. Compare two polling transmit strategies:

Strategy P: wait for `TXE` flag, then write to `DR`.

Strategy Q: write to `DR`, then wait for `TC` flag.

What is the key **side effect** of Strategy Q? [CO2/PO2]

- (A) The TC flag cannot be polled; it is only available via interrupt
- (B) Strategy Q returns before the byte reaches the shift register, causing data loss
- (C) Strategy Q blocks for an extra full byte-time *after* the data has left the shift register, unnecessarily reducing maximum throughput
- (D) Strategy Q skips the first byte of every new transmission burst

21. A ring buffer uses `BUF_SIZE=64` and indices `volatile uint8_t head=0, tail=0;`. The ISR increments `head`; main increments `tail`. How many bytes can the buffer hold **before it is full**? [CO2/PO2]

- (A) 62 bytes
- (B) 64 bytes
- (C) 63 bytes — one slot is reserved so the full condition `(head+1)%SIZE == tail` can be distinguished from empty
- (D) 32 bytes (half capacity is the recommended safe limit)

22. `GPIOA->MODER` is a 32-bit register with 2 bits per pin. What is the correct C expression to set `PA8` as Alternate Function mode without changing any other bits? [CO1/PO1]

- (A) `GPIOA->MODER &= ~(3UL<<16); GPIOA->MODER |= (2UL<<16);`
- (B) `GPIOA->MODER = (2UL << 16);`
- (C) `GPIOA->MODER |= (2UL<<8);`
- (D) `GPIOA->MODER &= ~(3UL<<8); GPIOA->MODER |= (2UL<<8);`

23. A student uses `HAL_UART_Receive_IT(&huart2, &rxByte, 1)` and re-arms it in the callback. At 115 200 baud one byte arrives every $\approx 87 \mu s$. The callback takes $200 \mu s$. What is the **most likely** consequence? [CO2/PO2]

- (A) One or more incoming bytes are lost while the callback executes, because `RXNE` fires again before `HAL_UART_Receive_IT` is re-armed
- (B) The next byte is safely stored in a hardware double-buffer inside the USART peripheral
- (C) HAL automatically queues up to 16 bytes in the USART FIFO
- (D) The MCU resets due to a watchdog timeout triggered by the long callback

24. The User Button B1 is on PC13. A student configures EXTI line 13 for falling-edge detection but **omits the write to** `SYSCFG->EXTICR[3]`. What is the default EXTI13 source, and what is the practical effect? [CO2/PO2]

- (A) EXTI13 has no default and is disabled until SYSCFG is explicitly written
- (B) Default source is Port A (PA13); button presses on PC13 will *not* trigger the ISR
- (C) Default source is Port C; the ISR fires correctly even without the SYSCFG write
- (D) Default source is Port B; the ISR fires on PB13 events instead

25. USART2 ISR has NVIC preemption priority **0**; EXTI15_10 ISR has preemption priority **2**. The User Button is pressed while the USART2 ISR is executing. What happens? [CO2/PO2]

- (A) The EXTI15_10 ISR immediately preempts USART2 ISR because button events are hardware-urgent
- (B) Both ISRs execute simultaneously on the dual-issue Cortex-M4 pipeline
- (C) A fault occurs because two ISRs cannot be pending simultaneously in the NVIC
- (D) The EXTI15_10 ISR is pended and waits until the USART2 ISR completes, because a higher priority number means lower preemption ability on Cortex-M4

26. A Dhaka-based startup deploys IoT air-quality sensors using STM32 nodes. Each node must transmit 200-byte packets every 500 ms over UART to a LoRa radio while also sampling sensors. Which firmware architecture **best** keeps the CPU free for sensor work? [CO3/PO6]

- (A) DMA TX in normal mode triggered once per 500 ms; CPU enters sleep between transfers
- (B) Polling TXE in a busy-wait loop for each of the 200 bytes
- (C) Bit-bang UART implemented entirely inside a SysTick ISR
- (D) UART TX interrupt per byte with a transmit ring buffer drained by the TXEIE ISR

27. Which single C statement **correctly** clears the EXTI pending flag for line 13 inside an ISR? (The EXTI PR register is write-1-to-clear.) [CO2/PO2]

- (A) `EXTI->PR = (1UL<<13);`
- (B) `EXTI->PR &= ~(1UL<<13);`
- (C) `EXTI->IMR &= ~(1UL<<13);`
- (D) `NVIC_ClearPendingIRQ(EXTI15_10_IRQn);`

28. A student must configure PA5 as a general-purpose push-pull output. Which statement about the required register writes is **correct**? [CO1/PO1]

- (A) Both `MODER[11:10]=01` and `OTYPER[5]=1` must be written
- (B) `MODER[11:10]=01` must be written; the reset defaults of OTYPER (push-pull) and PUPDR (no pull) are already correct
- (C) Both `MODER[11:10]=10` and an AFR entry must be written
- (D) Only `ODR[5]=1` needs to be set to activate the output

29. Consider the ISR fragment below: [CO2/PO2]

```
void USART2_IRQHandler(void) {
    char c = (char)USART2->DR;
    ring[head] = c;
    head = (head + 1) % BUF_SIZE;
}
```

A student notices that the ORE flag occasionally freezes the UART receive. What **must** be added to this ISR to fix the problem?

- (A) Increase `BUF_SIZE` to 256 to prevent overrun
- (B) Check the ORE flag and clear it by reading `USART2->DR` when set
- (C) Replace the modulo operation with a bitmask for execution speed
- (D) Call `NVIC_DisableIRQ(USART2_IRQn)` at the top of the ISR

30. DMA1 Stream 5 is configured for USART2 RX in **circular mode** with `NDTR=64` and `rxBuf[64]`. After 64 bytes are received, what does the DMA controller do **automatically**? [CO2/PO2]

- (A) It resets NDTR to 64, rewinds M0AR to `rxBuf[0]`, raises TC, and continues receiving
- (B) It raises a DMA configuration error because circular mode cannot be used with a USART peripheral
- (C) It raises a TC interrupt and halts; the CPU must re-enable the stream manually
- (D) It switches to M1AR (double-buffer mode) automatically

— End of Question Paper —

Total Questions: 30 | Total Marks: 30 | Submit this paper with your Answer Sheet

UNIVERSITY OF DHAKA

Department of Computer Science and Engineering

CSE 2206 — Microcontroller and Embedded System

MCQ ANSWER SHEET SET C

2nd Year, 2nd Semester 2024 | Full Marks: 30 | Time: 40 Minutes

Student Name: Student ID: Section/Group:

Date: Exam Hall: Seat No.:

Instructions: Shade **one** bubble per question using a dark pen or pencil. Do not use ticks, crosses, or circles. Write your answers *here only* — not on the question paper. Erased or ambiguous answers score zero.

Answer Grid — Shade the correct bubble

Q#	A	B	C	D	Q#	A	B	C	D
1	☐	☐	☐	☐	16	☐	☐	☐	☐
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3	☐	☐	☐	☐	18	☐	☐	☐	☐
4	☐	☐	☐	☐	19	☐	☐	☐	☐
5	☐	☐	☐	☐	20	☐	☐	☐	☐
6	☐	☐	☐	☐	21	☐	☐	☐	☐
7	☐	☐	☐	☐	22	☐	☐	☐	☐
8	☐	☐	☐	☐	23	☐	☐	☐	☐
9	☐	☐	☐	☐	24	☐	☐	☐	☐
10	☐	☐	☐	☐	25	☐	☐	☐	☐
11	☐	☐	☐	☐	26	☐	☐	☐	☐
12	☐	☐	☐	☐	27	☐	☐	☐	☐
13	☐	☐	☐	☐	28	☐	☐	☐	☐
14	☐	☐	☐	☐	29	☐	☐	☐	☐
15	☐	☐	☐	☐	30	☐	☐	☐	☐

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CO1 (PO1)	CO2 (PO2)	CO3 (PO6)	Total /30	Percentage
/8	/18	/4	/30	%

Examiner Name: **Signature:**

ANSWER KEY — Examiner Copy Only. Remove before printing student sheets.

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans		
1	D	2	B	3	A	4	D	5	B	6	B
7	C	8	C	9	A	10	D	11	C	12	A
13	B	14	C	15	C	16	C	17	B	18	C
19	A	20	C	21	C	22	A	23	A	24	B
25	D	26	A	27	A	28	B	29	B	30	A

— End of Answer Sheet —

CSE 2206 — MASTER ANSWER KEY

MCQ Laboratory Test — All Three Sets (A, B, C)

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Q#	Set A	Set B	Set C	Q#	Set A	Set B	Set C
1	A CO2	C CO3	D CO2	16	A CO3	A CO2	C CO2
2	C CO2	B CO1	B CO2	17	C CO1	A CO2	B CO3
3	D CO2	B CO2	A CO1	18	C CO2	B CO2	C CO1
4	A CO3	A CO2	D CO2	19	A CO2	A CO3	A CO1
5	C CO2	A CO1	B CO1	20	B CO1	D CO2	C CO2
6	C CO1	B CO2	B CO2	21	A CO2	D CO2	C CO2
7	C CO1	A CO1	C CO3	22	D CO3	A CO1	A CO1
8	B CO2	D CO2	C CO1	23	B CO1	D CO2	A CO2
9	B CO2	D CO2	A CO2	24	A CO2	A CO2	B CO2
10	D CO2	A CO2	D CO3	25	C CO1	A CO1	D CO2
11	A CO2	B CO3	C CO2	26	B CO2	D CO1	A CO3
12	C CO2	B CO2	A CO2	27	A CO2	C CO2	A CO2
13	C CO3	D CO2	B CO2	28	B CO1	C CO2	B CO1
14	A CO2	A CO2	C CO1	29	A CO1	C CO1	B CO2
15	D CO2	A CO3	C CO2	30	A CO2	B CO1	A CO2

Original Question Cross-Reference (where each original Q appears in each set)

Orig	Set A	Set B	Set C	Orig	Set A	Set B	Set C	Orig	Set A	Set B	Set C
Q1	Q28(B)	Q22(A)	Q14(C)	Q11	Q3(D)	Q27(C)	Q20(C)	Q21	Q30(A)	Q18(B)	Q21(C)
Q2	Q17(C)	Q5(A)	Q8(C)	Q12	Q9(B)	Q23(D)	Q13(B)	Q22	Q14(A)	Q10(A)	Q29(B)
Q3	Q20(B)	Q30(B)	Q28(B)	Q13	Q8(B)	Q21(D)	Q4(D)	Q23	Q5(C)	Q3(B)	Q1(D)
Q4	Q29(A)	Q7(A)	Q19(A)	Q14	Q18(C)	Q28(C)	Q6(B)	Q24	Q27(A)	Q16(A)	Q23(A)
Q5	Q23(B)	Q2(B)	Q22(A)	Q15	Q2(C)	Q8(D)	Q2(B)	Q25	Q24(A)	Q6(B)	Q11(C)
Q6	Q7(C)	Q29(C)	Q3(A)	Q16	Q10(D)	Q24(A)	Q24(B)	Q26	Q12(C)	Q14(A)	Q30(A)
Q7	Q6(C)	Q26(D)	Q18(C)	Q17	Q15(D)	Q20(D)	Q12(A)	Q27	Q4(A)	Q19(A)	Q26(A)
Q8	Q25(C)	Q25(A)	Q5(B)	Q18	Q21(A)	Q12(B)	Q25(D)	Q28	Q16(A)	Q1(C)	Q7(C)
Q9	Q26(B)	Q13(D)	Q16(C)	Q19	Q19(A)	Q4(A)	Q27(A)	Q29	Q22(D)	Q11(B)	Q17(B)
Q10	Q11(A)	Q9(D)	Q15(C)	Q20	Q1(A)	Q17(A)	Q9(A)	Q30	Q13(C)	Q15(A)	Q10(D)

— End of Master Answer Key —